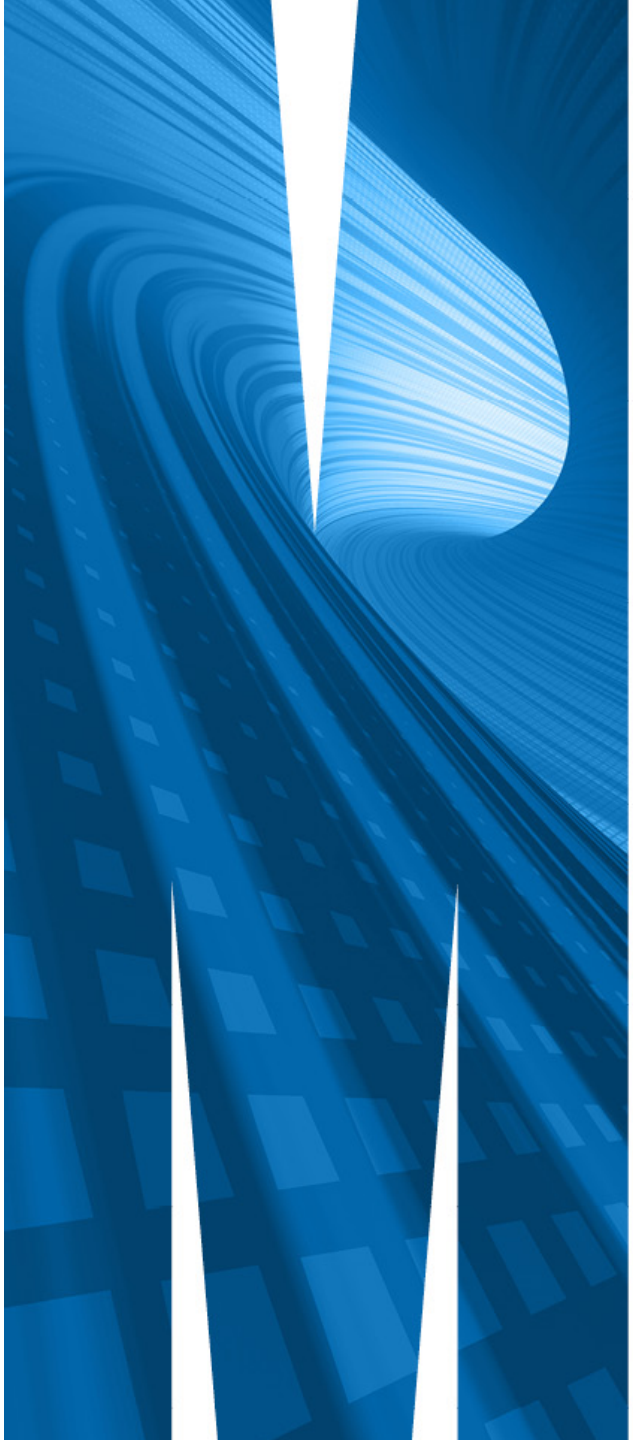
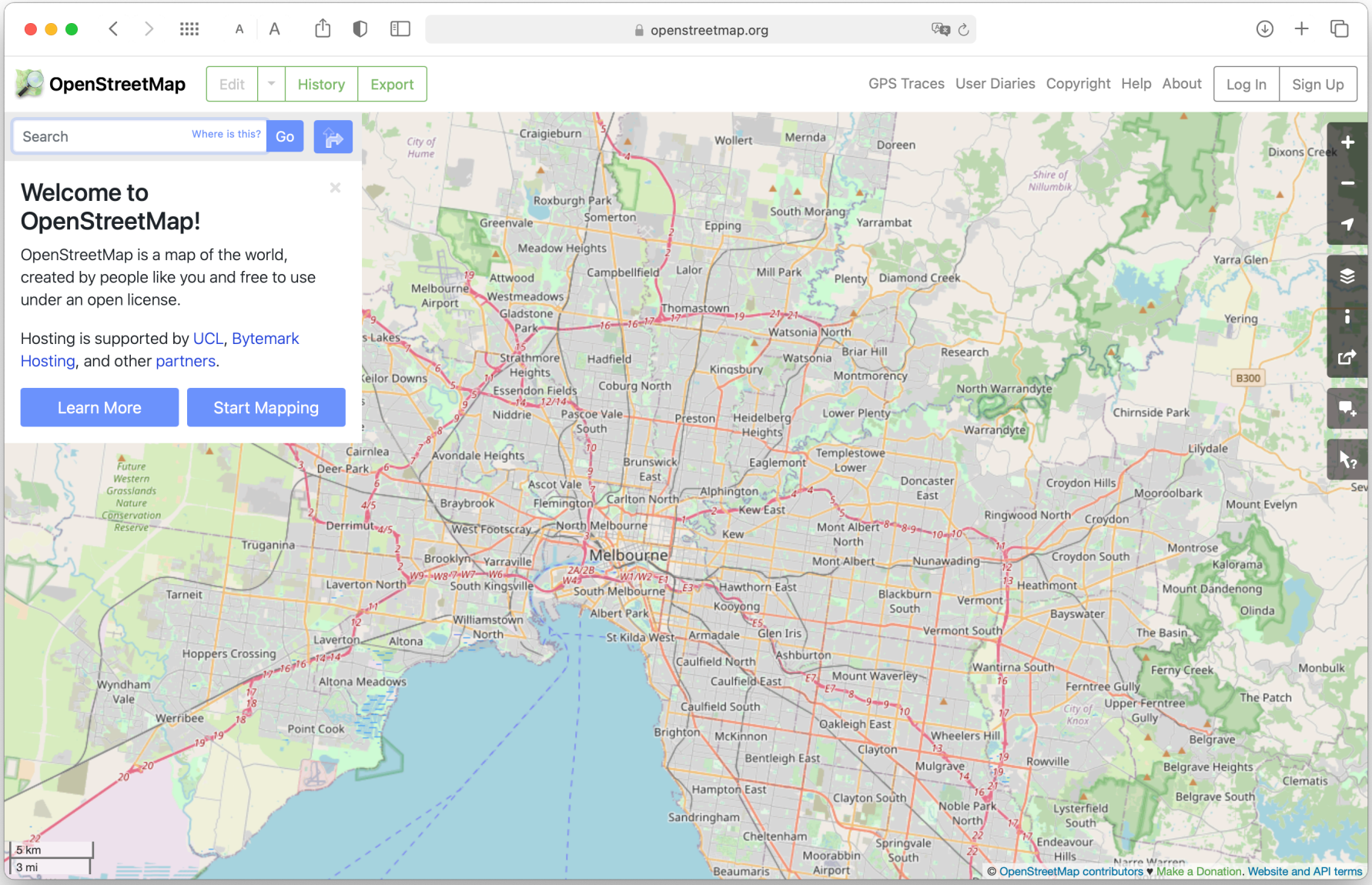
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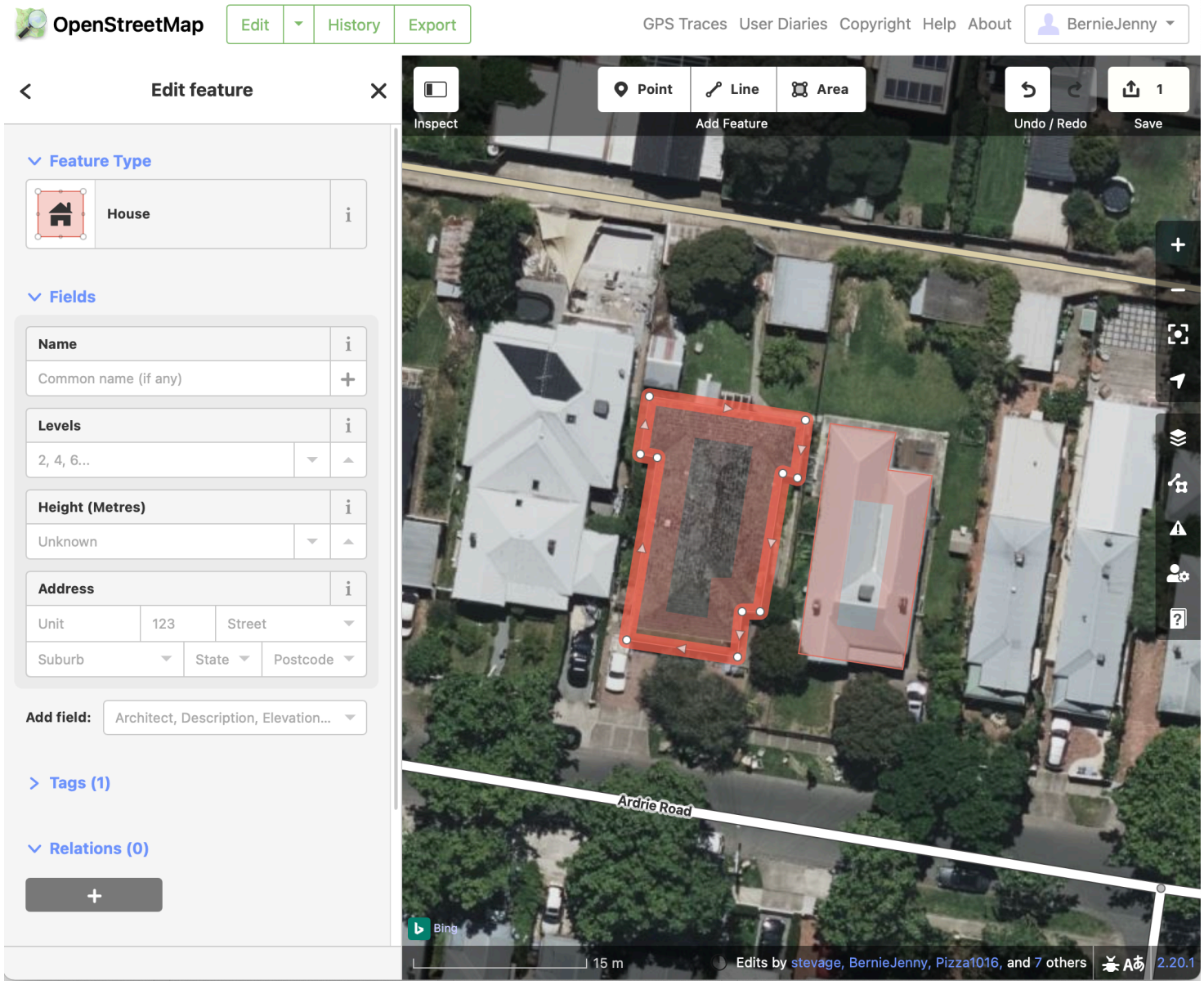


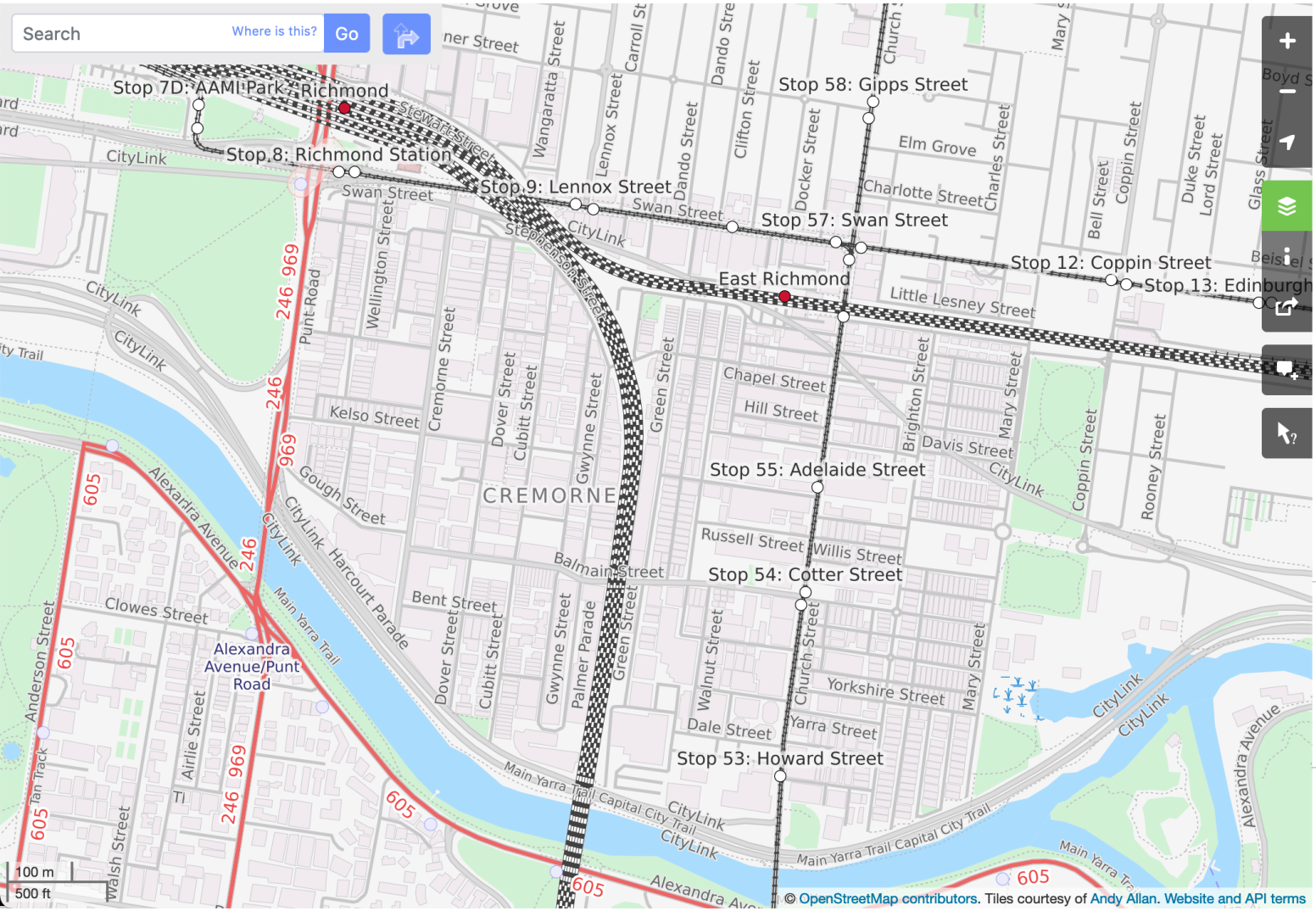
FIT2179 Data Visualisation

Week 7: Tiled web maps



OpenStreetMap.org


OpenStreetMap
Edit
History
Export
GPS Traces
User Diaries
Copyright
Help
About
BernieJenny
Edit feature
Feature Type
House
Fields
Name
Common name (if any)
Levels
2, 4, 6...
Height (Metres)
Unknown
Address
Unit
123
Street
Suburb
State
Postcode
Add field:
Architect, Description, Elevation...
Tags (1)
Relations (0)
Inspect
Point
Line
Area
Add Feature
Undo / Redo
1
Save
Ardrie Road
Bing
15 m
Edits by stevage, BernieJenny, Pizza1016, and 7 others
2.20.1



OpenStreetMap data visualised in various styles
Examples: OpenCycleMap, Toner, Terrain, Watercolor



OpenStreetMap data visualised in various styles
Examples: OpenCycleMap, Toner, Terrain, Watercolor

Tiled Web Maps or “Slippy Maps”

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- Mercator projection with latitude limits at 85° to create a square map
- 256×256 pixels per tile, PNG, JPEG or similar format
- Zoom level 0 entire world in a single tile
- Increasing zoom level by 1: each tile is replaced by 4 tiles
- Level 1 has 2×2 tiles
  Level 2 has 4×4 tiles
  Level n has $2^n \times 2^n$ tiles
- Levels up to 22 or 23



Leaflet

Leaflet

an open-source JavaScript library
for mobile-friendly interactive maps

Overview Tutorials Docs Download Plugins Blog

Web map libraries load and render tiles, combine tile layers and GeoJSON/ TopoJSON features, and provide interactive tools.

http://leafletjs.com or
https://openlayers.org

May 18, 2023 — Leaflet 1.9.4 has been released!

Leaflet is the leading open-source JavaScript library for mobile-friendly interactive maps. Weighing just about 42 KB of JS, it has all the mapping features most developers ever need.

Leaflet is designed with *simplicity*, *performance* and *usability* in mind. It works efficiently across all major desktop and mobile platforms, can be extended with lots of plugins, has a beautiful, easy to use and well-documented API and a simple, readable source code that is a joy to contribute to.

